SUPERVISOR MEETING

Literature Review & Research Gap Validation

Adaptive Task Offloading in the IoT–Edge–Cloud Continuum

Eight supplied studies + our current iFogSim2 implementation



Main conclusion: the broad algorithmic gap is not confirmed yet; the next contribution must be justified against prior work and tested fairly.

What the supervisor asked us to deliver

The presentation is focused on the literature review, the comparison table, and a defensible plan for what comes next.

Required from the literature review

- Compare the eight supplied papers against our project.
- For every study, extract methodology, dataset/workload, metrics, and limitations.
- Identify which parts of our idea already exist in published work.
- Define a contribution we can defend and test in iFogSim2—not just combine known algorithms.

Current status

DONE (INITIAL)	Eight source records organized; matrix drafted.
PARTIAL	Some papers need full-text / publisher-metadata confirmation; search log is not systematic yet.
NOT PROVEN	No distinct algorithmic novelty is established.
NEXT DECISION	Supervisor chooses empirical contribution vs. new algorithm.

Basis: supplied papers, literature matrices, research-gap audit v29, technical audit, and v26 results bundle.

Our current implementation: the comparison baseline

Describe what exists today separately from what is only proposed.

METHOD / ACTIONS

iFogSim2 + Python JSONL integration. Current v26 reports a tabular Q-value policy with 81 states and γ=0; functionally a contextual-bandit-style decision policy, not validated sequential MDP Q-learning. Actions: Local / Edge / Cloud.

WORKLOAD / SCENARIOS

Synthetic generated tasks with task-class deadlines S/M/L = 20/50/120 simulation units. Two v26 scenarios: heterogeneous deadlines; Edge-1 compute-service unavailable for new assignments during t=250–500, then recovery.

METRICS AVAILABLE

Mean, P95 and P99 latency; deadline-violation rate (overall and by task class); completion rate; task counts and orphaned-task accounting. Decision overhead, full energy and true CPU service-start time are not validated as primary metrics.

LIMITATIONS / OVERLAP

No public offloading trace; current results cover 5 test seeds per scenario. Recent work already covers three-tier DQN, deadline-aware RL/bandits, dynamic load, and partial offloading. Older audit configurations differ and must not be merged with v26.

Version control rule: v25 runs, v26 (60 runs), and legacy Seed 1/2 audits are separate evidence sets.

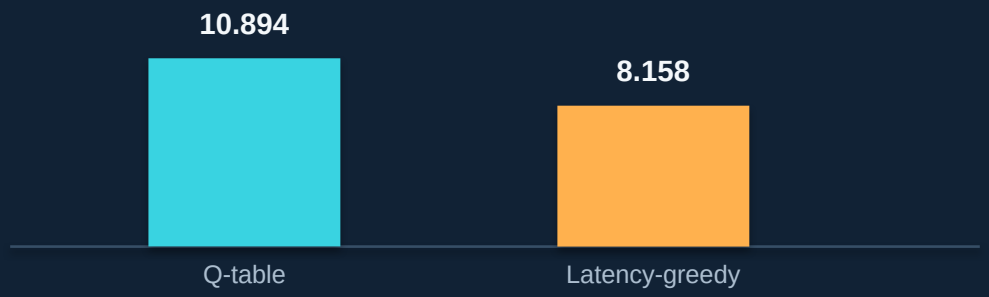
Source: v26 results README + manifest/metrics; Member 3 technical audit. Simulation-time units are not assumed to be milliseconds.

What the current v26 results actually show

60 runs = 2 scenarios × 6 policies × 5 test seeds; the values below are means across five seed-level means.

HETEROGENEOUS DEADLINES

Mean end-to-end latency (simulation units)

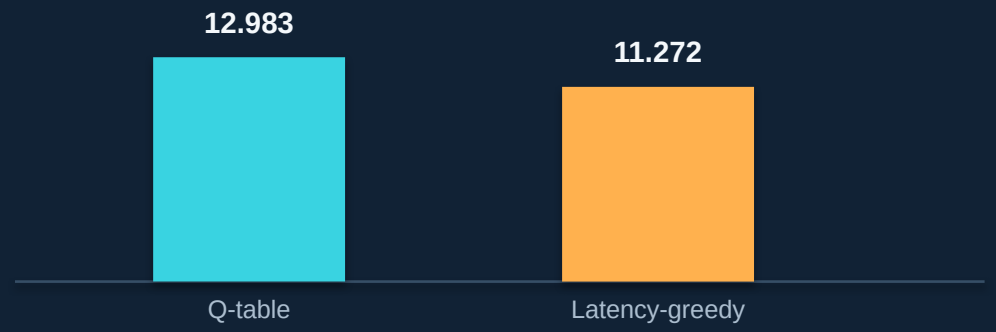


Paired t-test (exploratory, n=5): p = 0.0004449

Deadline misses: 0.000% vs 0.000% (Q-table vs greedy)

EDGE-1 SERVICE OUTAGE / RECOVERY

Mean end-to-end latency (simulation units)



Paired t-test (exploratory, n=5): p = 0.0002422

Deadline misses: 0.209% vs 0.141% (Q-table vs greedy)

Interpretation: Latency-greedy is faster in both v26 scenarios. These five-seed comparisons do not establish a generally superior learned policy; the p-values are exploratory and should be reported cautiously.

Source: Results/README_results.txt + metrics.csv + paired differences. “Deadline-greedy” is identical to latency-greedy by construction; do not count it as an independent algorithm.

Papers 1–2 | Adaptive RL and real-time smart-city orchestration

Both papers learn state-dependent placement, but differ in system topology and evaluation setting.

STUDY / SOURCE	METHODOLOGY	DATASET / WORKLOAD	METRICS	LIMITATIONS / OVERLAP
P1 • RL-Based Adaptive Task-Offloading in Mobile-Edge Computing for IoT Networks **IEEE IoT Magazine • DOI 10.1109/MIOT.2025.3623451**	Adaptive RL for cellular MEC; abstract claims multi-agent DQN / safe RL, while the detailed method section describes tabular Q-learning. Includes NOMA, server selection and transmit-power decisions.	Synthetic/simulated Poisson arrivals; roughly 10–100 active nodes and five edge servers. No public task/offloading trace identified.	Latency, energy, QoS/reliability and throughput/user-satisfaction measures.	Algorithm description is internally inconsistent; setup units need checking; no reproducible public trace/raw outputs. Cellular NOMA scope differs from our current iFogSim2 model.
P2 • Edge-Cloud Adaptive Framework for Real-Time IoT Data Processing in Smart Cities **IEEE ACROSET 2025 • DOI 10.1109/ACROSET66531.2025.11280634**	Double Dueling DQN at edge gateways; Edge vs Cloud inference per micro-batch; MQTT telemetry and a cloud-side fairness controller.	102 traffic-camera streams + 64 air-quality sensors, with synthetic flash-crowd bursts; Jetson Nano + AWS EKS testbed. Public dataset link/licence not verified in the supplied PDF.	Mean/P95 end-to-end latency, 100-ms SLO compliance, upstream bandwidth, energy/frame, Jain GPU fairness and control overhead.	Only Edge/Cloud actions, not three-tier Local/Edge/Cloud; primary baselines are Cloud-only and static Fog; camera resolution wording varies across sections.

Evidence note: paper-specific details are summarized from the supplied PDFs/matrix; unclear or inconsistent reporting is marked rather than inferred.

Papers 3–4 | Three-tier DQN and the broader survey

One is the closest algorithmic overlap; the survey is useful context but is not a direct benchmark.

STUDY / SOURCE	METHODOLOGY	DATASET / WORKLOAD	METRICS	LIMITATIONS / OVERLAP
P3 • Deep Q-Network based Task Offloading Optimisation for Edge-IIoT System **IEEE ICICIT 2026 • DOI 10.1109/ICICIT69063.2026.11634047**	DQN with replay/target-network elements; LSTM is also described. Three actions: Local, Edge, Cloud. Reward is described in terms of latency, energy, cost and deadline violations.	Mentions Kaggle Edge-IIoT network traffic; simulation also uses fixed task/network parameters. The mapping from network records to compute cycles, deadlines and offloading ground truth is not reproducibly explained.	Latency, energy, execution cost, reward and completion-before-deadline; the paper reports 94.56% deadline completion.	Very high overlap in three-tier actions and deadline-aware RL. Equations, LSTM–DQN integration, baseline parity and seed protocol need verification; treat reported numbers as author-reported.
P4 • Deep Learning-Centric Task Offloading in IoT-Fog-Cloud Continuum: A State-of-the-Art Review **IEEE Access 2025 • DOI 10.1109/ACCESS.2025.3599190**	Survey/taxonomy of DL/RL offloading methods, architecture types, simulation frameworks, metrics and open issues; no single executable policy.	The reviewed literature is the evidence base; stated article window is 2010–2023. No single runtime dataset or workload.	Maps latency, energy, network/resource utilization, cost and other QoS measures across papers; no single benchmark result.	Not a direct policy baseline; its review window ends in 2023, so it cannot establish a 2025–2026 gap alone. Requires an up-to-date targeted review.

Evidence note: paper-specific details are summarized from the supplied PDFs/matrix; unclear or inconsistent reporting is marked rather than inferred.

Papers 5–6 | Deadline-aware RL and policy benchmarking

One overlaps on deadlines; the other helps us design a broader, context-dependent baseline evaluation.

STUDY / SOURCE	METHODOLOGY	DATASET / WORKLOAD	METRICS	LIMITATIONS / OVERLAP
P5 • An Adaptive Edge-Cloud Task Offloading Framework using RL for Latency-Sensitive AI Applications **IEEE ICOECA 2026 • DOI 10.1109/ICOECA68095.2026.11485334**	MDP-based RL; Edge vs Cloud only. State includes task characteristics, edge queue/utilization, bandwidth and cloud queue. Reward combines latency with deadline satisfaction.	Synthetic profiles for vision, AR/VR, health-IoT and industrial CPS; no named public trace or code link identified in the supplied paper.	Mean/P95 latency, deadline success, SLA violations, energy/task and robustness under bandwidth/workload changes.	Deadline-aware adaptive RL and P95 are already covered. Not Local/Edge/Cloud; article terminology/equations and some reported claims are internally unclear and need caution.
P6 • An Analysis of Task Offloading Approaches in Edge-Cloud Continuum **IEEE COMPSAC 2025 • DOI 10.1109/COMPSAC65507.2025.00206**	EdgeCloudSim comparison of five rule/fuzzy policies: NETWORK_BASED, UTILIZATION_BASED, FUZZY_BASED, FUZZY_COMPETITOR and HYBRID; five configurations and ten iterations.	Synthetic sample_app4 workloads: AR, health, heavy-compute and infotainment; moderate and high device loads. No external task trace identified.	Failure ratio, average service time, task counts/completions, placement distribution, and per-app/device outcomes.	No learning-based baseline; synthetic workload; some scenario/device counts vary in reporting; paired-seed effects and uncertainty intervals are not clearly documented.

Evidence note: paper-specific details are summarized from the supplied PDFs/matrix; unclear or inconsistent reporting is marked rather than inferred.

Papers 7–8 | Partial offloading and the CAV survey

Partial task splitting is already published; the CAV survey is adjacent context rather than a direct comparator.

STUDY / SOURCE	METHODOLOGY	DATASET / WORKLOAD	METRICS	LIMITATIONS / OVERLAP
P7 • Hierarchical DRL-Based Partial Task Offloading in Device-Edge-Cloud Computing Architecture **IEEE TCE 2026 • DOI 10.1109/TCE.2025.3611973**	Hierarchical DRL: high-level Double DQN selects tiers/actions; low-level PPO chooses task split ratios across device, Edge and Cloud.	Synthetic task size, CPU-cycle and tolerable-latency profiles over two network settings; authors' Python/PyTorch simulation; no public offloading trace identified.	Task acceptance ratio, energy, resource utilization, episode reward, latency constraints and generalization to new task profiles.	Partial offloading is not a new contribution by itself; requires divisible-task and split/communication-overhead models. Different simulator and higher implementation complexity.
P8 • Task Offloading for CAVs Edge Computing Environment: Taxonomy, Critical Review, and Future Road Map **ACM Computing Surveys 2026 • DOI 10.1145/3783984**	Survey/taxonomy of Connected and Autonomous Vehicle offloading, mobility, architectures, methods, datasets and simulators; no single policy or reward.	Survey evidence; discusses KITTI, Waymo Open, BDD100K and ApolloScape for automotive perception. These are not aligned per-task offloading outcome traces.	Summarizes metrics used across CAV literature—latency, reliability, mobility/resource factors; no single benchmark metrics/results.	CAV-specific and not a direct baseline. Vision datasets do not directly provide task CPU cycles, queues, deadlines and execution outcomes needed by our simulator.

Evidence note: paper-specific details are summarized from the supplied PDFs/matrix; unclear or inconsistent reporting is marked rather than inferred.

The key comparisons: what is already covered?

Our project must be compared against the literature at the mechanism level—not only by title or simulator.

RESEARCH ELEMENT	EXISTING EVIDENCE IN THE 8 STUDIES	OUR CURRENT POSITION	WHAT THIS MEANS
Three-tier Local / Edge / Cloud	P3 uses Local/Edge/Cloud; P7 distributes subtasks across device/Edge/Cloud.	Current policy chooses Local / Edge / Cloud.	Three-tier decision-making alone is not a gap.
Deadline-aware placement	P3 and P5 use task deadlines; other closely related work in the extended search also covers deadline-aware bandits/gates.	v26 assigns class-level deadlines S/M/L = 20/50/120 units.	Deadline awareness alone is insufficient as novelty.
Dynamic load / network	P2 uses flash-crowd bursts; P5 varies workloads/bandwidth; P6 varies load and bandwidth.	v26 has heterogeneous deadlines and Edge-1 compute-service outage/recovery.	Need a sharper test protocol and outcome than simply “dynamic conditions”.
Risk, outage, fallback	Several papers address reliability, constraint handling or dynamic conditions; LITO/O2C/E2DA appear in the extended scan.	Current outage experiment blocks new assignments to Edge-1; it is not a physical host crash.	Define fault model and report what is actually measured.
Partial offloading	P7 explicitly uses hierarchical DDQN + PPO task splitting.	Current project performs whole-task placement, not split ratios.	Partial offloading is not new by itself and adds major modeling effort.
Reproducibility / trace data	Several supplied studies use synthetic workloads; public task-level traces and matched seed protocols are limited or unclear.	Current experiments use synthetic v26 workload and 5 test seeds per scenario.	A transparent, paired and fully documented benchmark is a plausible methodological direction—not automatic algorithmic novelty.

Research-gap verdict: no new algorithmic gap is confirmed yet. E2DA, LITO, O2C/CCMAB and forecast-driven LinUCB need close follow-up before any novelty claim.

Supplementary related-work leads from the targeted search are not replacements for full-text verification: E2DA, LITO, O2C/CCMAB, forecast-driven LinUCB and H-DQAC.

What can we add that is testable and defensible?

Candidate direction for supervisor approval—not a claim that novelty has already been established.

Candidate contribution

A reproducible three-tier evaluation of lightweight offloading policies under controlled operating-regime changes.

- Paired comparisons: current Q-table/bandit, latency-greedy, load-aware heuristic, Random, Local-only, Edge-only and Cloud-only. DQN only if feasible and specified fairly.
- Same task stream and seed per policy; frozen policy at test time; separate train and test seeds; publish config, scenario file, code version and aggregation script.
- Scenarios: heterogeneous deadlines, bandwidth/network changes, Edge service unavailability/recovery, and—if time permits—repeated A→B→A regime shifts.
- Primary metrics: mean/P95/P99 end-to-end latency, deadline misses by task class, completion and pending counts, recovery time and cumulative misses after a change.
- Report decision overhead and traffic bytes only if instrumentation is validated. Do not claim energy savings or measured CPU utilization without a trustworthy model.

How it could add value

- A clean, same-simulator comparison, not an incomparable cross-paper score table.
- Separate steady-state performance from transient behavior after a regime change.
- Explicit task accounting, units, uncertainty intervals and failure definitions.
- Honest open artifacts: scenario configs, seeds, logs and result aggregation.

Novelty still requires a fresh overlap check.

Important: dynamic-shift testing, deadline awareness and lightweight bandits already have prior art. The exact differential must be verified against the closest recent papers.

Work plan: from the current matrix to a defensible study

Do not add a new algorithm until the evidence, metrics and protocol are pinned down.

P0

Evidence & definitions

- Member 1: verify title/venue/DOI and extract methodology, workload, metrics and limitations for each of the eight papers from full text.
- Member 2: finish search log (queries, date, results, duplicates, inclusion/exclusion reasons) for IEEE Xplore, SpringerLink and ScienceDirect.
- Mark + Member 3: reconcile v25/v26/legacy configs; freeze code/model/config hash; validate simulation-time units and E2E definition.

P1

Fair experiments

- Freeze policies and definitions; remove duplicate deadline-greedy from the independent baseline count.
- Use identical workload per seed across policies; keep training seeds separate from test seeds; record every task terminal state.
- Run a pilot, then target 20+ paired test seeds if feasible; report means, spread, 95% CIs and paired differences.

P2

Decide & report

- Analyze task class, regime, latency tail and deadline miss behavior; separate steady-state from recovery performance.
- Re-check novelty against E2DA/LITO/O2C/forecast-driven LinUCB and other closest recent work.
- Write contribution only after supervisor agrees on algorithmic novelty vs. reproducible empirical study.

Suggested formal run target: 20+ paired test seeds if compute/time permit; retain 5-seed v26 results as preliminary, not definitive.

Decisions needed before we commit to the next phase

Use the meeting to agree on the contribution and the practical evaluation requirements.

Questions to decide

- Does the project require a genuinely new algorithm, or is a reproducible experimental comparison acceptable?
- Can we freeze iFogSim2 as the primary simulator and approve the current scenario scope?
- Which baselines and minimum metrics are required for the formal evaluation?
- Is synthetic workload generation acceptable if assumptions and seeds are documented, or is a mapped trace required?
- What evidence will the supervisor accept as sufficient to validate the Research Gap?

Eight papers in the comparison

P1 RL-Based Adaptive Task-Offloading… — DOI 10.1109/MIOT.2025.3623451

P2 Edge-Cloud Adaptive Framework… — DOI 10.1109/ACROSET66531.2025.11280634

P3 Deep Q-Network… Edge-IIoT — DOI 10.1109/ICICIT69063.2026.11634047

P4 Deep Learning-Centric Task Offloading Review — DOI 10.1109/ACCESS.2025.3599190

P5 Adaptive Edge-Cloud RL Framework — DOI 10.1109/ICOECA68095.2026.11485334

P6 Analysis of Task Offloading Approaches — DOI 10.1109/COMPSAC65507.2025.00206

P7 Hierarchical DRL Partial Offloading — DOI 10.1109/TCE.2025.3611973

P8 CAV Task Offloading Survey — DOI 10.1145/3783984

Recommended meeting position: the gap is not yet proven; show the evidence, ask for scope approval, and agree on the validation protocol.